Time Decay in Fellegi–Sunter Record Linkage: A Literature Survey

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Abstract

This survey reviews the documented line of work that treats comparison evidence in probabilistic (Fellegi–Sunter) record linkage as *decaying* with the gap between two records. The static m/u parameters of the classical model have, since 2011, been extended with temporal variants in which agreement probabilities, comparison weights, or the final similarity score become functions of the elapsed time between record timestamps. We summarize the primary points of the founding temporal-linkage papers and of the works closest to the present project: Li et al. (2011), Christen and Gayler (2013), Chiang et al. (2014), Hu et al. (2017), and Shim's *TimeLink* (2026). We then present an information-theoretic reading: a Fellegi–Sunter weight is a log-likelihood ratio measured in bits, its match-conditional expectation is a Kullback–Leibler divergence, and u is tied, in the frequency model, to a Rényi collision entropy; "decay of evidence with age" is therefore the statement that the mutual information between a comparison outcome and match status falls to zero over time. The survey closes with a gap analysis of what, to our knowledge, remains unpublished.

1 Where decay can enter Fellegi–Sunter

A Fellegi–Sunter (FS) linkage scores a candidate pair of records (a, b) by summing, over comparison fields c and comparison levels ℓ_c, the log-likelihood ratio

$$w_c(\ell_c) \;=\; \log_2 \frac{P(\ell_c \mid M)}{P(\ell_c \mid U)} \;=\; \log_2 \frac{m_c(\ell_c)}{u_c(\ell_c)}, \tag{1}$$

where M is "same entity," U is "different entity," m_c is the probability of agreement given a true match, and u_c is the probability of agreement given a non-match (the coincidence rate). Posterior mass is monotone in the summed weights [1, 7]. FS assumes conditional independence across fields and, crucially for this survey, that m_c and u_c are *constants*, independent of any covariate of the pair.

The temporal literature inserts "decay" at four distinct points:

1. **Blocking and similarity.** The retrieval score is attenuated by the timestamp gap (Li et al. 2011; the present project's multi-view blocker).
2. **Weight exponent.** The per-field contribution $\log(m_c/u_c)$ is multiplied by a decay factor of the gap.
3. **Field probability m_c.** The agreement probability itself becomes a function $m_c(\Delta t)$, converging towards the coincidence baseline as the gap grows; equivalently $P(\gamma_c \mid M, \Delta t) \to P(\gamma_c \mid U)$. This is where "decay m, keep u fixed" sits (TimeLink; the present project).

4. **Value transition models.** m and u are replaced by a full distribution over what a value becomes after Δt (Chiang et al. 2014; Li et al. 2015; TimeLink's semi-Markov models).

2 Primary sources, chronologically

2.1 Li, Dong, Maurino, Srivastava (2011): first explicit time decay

Pei Li, Xin Luna Dong, Andrea Maurino, Divesh Srivastava. *Linking temporal records.* PVLDB 4(11):956–967, 2011. DOI: https://doi.org/10.14778/3402707.3402733 [2].

This is the paper that defined temporal record linkage: records carry a time-stamp and the task is to identify records that describe the same entity over time. Two mechanisms are proposed:

- *Time decay*: "We apply time decay to capture the effect of elapsed time on value evolution" — the first explicit use of the term in the linkage literature.
- *Clustering rather than pairwise decisions*: "Instead of comparing each pair locally, we propose clustering methods that consider the order of records to make global decisions."

Experiments show the algorithms "significantly outperform traditional record linkage techniques on various temporal data sets." This paper establishes the vocabulary ("time decay") and the two axes — per-pair weight decay and global/orderings-aware decisions.

2.2 Christen and Gayler (2013): adaptive temporal entity resolution

Peter Christen, Ross W. Gayler. *Adaptive temporal entity resolution on dynamic databases.* PAKDD 2013 (LNAI 7819), pages 558–569. DOI: https://doi.org/10.1007/978-3-642-37456-2_47 [3].

The first work, to our knowledge, to make the effective m-probabilities explicitly gap-dependent and updated from data.

- Streaming setup: query records are matched against a growing database; a matched query is inserted back into the database (online, adaptive).
- For each attribute and time-lag bin the system maintains agree/disagree counts from accepted matches, yielding a data-driven agreement probability $P(A_{\Delta t} \mid S)$; the disagreement probability for non-matches is frequency-based ($1 - P_j(v)$, with P_j the value frequency).
- Each attribute similarity s_j is rescaled by a time-dependent weight: an agreement is down-weighted to the extent that different entities plausibly share the value; a disagreement is down-weighted to the extent that same entities plausibly changed it.
- Models fitted "adaptively" (update per insertion) and "non-adaptively" (frozen training-time counters). On synthetic data with noise the adaptive model wins; on the NC Voter database (about 2.4M records) the *adaptive* variant underperformed its frozen baseline, because with few true matches per query the counters are noisy — a cautionary result directly relevant to the calibration paradox seen in the white paper.

2.3 Chiang, Doan, Naughton (2014): entity evolution

Yue-Hsuan Chiang, AnHai Doan, Jeffrey F. Naughton. *Modeling entity evolution for temporal record matching.* SIGMOD 2014, pages 1175–1186. DOI: https://doi.org/10.1145/2588555.

2588560 [4].

The paper recognizes that "any such method relies at its heart on a model that captures information about how the entities evolve". It proposes:

- a more detailed model of the *reappearance* probability of an attribute value over time, conditionally dependent on past values, rather than a simple change-or-not decay;
- matching over *sets* of records, not pairs, for robustness;
- experimental evidence of improved accuracy and noise-resistance at minimal overhead.

This is the paper that turns "decay" into "stochastic value evolution," a full transition model; TimeLink's semi-Markov formulation is in this lineage.

2.4 Hu, Wang, Vatsalan, Christen (2017): regression classification

Yichen Hu, Qing Wang, Dinusha Vatsalan, Peter Christen. *Improving temporal record linkage using regression classification*. PAKDD 2017 (LNCS 10235), pages 561–573. DOI: https://doi.org/10.1007/978-3-319-57454-7_44 [5].

Verified against the accepted text. Key points:

- The decay probabilities of Li et al. are replaced by a *logistic regression* of the (time-gap) disagreement probability $\hat{d}_{\neq}(A, \Delta t)$, with support attributes (gender, age) and the gap as features; the regression replaces the (estimated) decay, while agreement decay $d_{=}$ is retained.
- The weight on attribute A becomes

  $$w_A(s_A, \Delta t) = 1 - s_A\, d_{=}(A, \Delta t) - (1 - s_A)\, d_{\neq}(A, \Delta t), \tag{2}$$

  with an *impact-reduction* coefficient α that damps the temporal blow (as $\alpha \to \infty$ the score re-approaches a non-temporal scorer).
- Evaluation on NC Voter subsets (5k–100k): at full strength the pure decayed model is poor; a moderate α and a larger training pool give the temporal signal a small win ("Mixed"). The paper explicitly reports that without the damping the decay models "degraded" the linkage quality.

Relevance to the project: this is the strongest published statement that a decayed m is not free; without value-frequency-aware u and a regularized (hard) gap effect it can pass below the untrained static baseline on real data — a result that the project's calibration experiments reproduce.

2.5 Shim (2026): TimeLink

Kwan Bo Shim. *TimeLink: time-aware record linkage with semi-Markov dynamics*. SSRN preprint 6468518, 2026. DOI: https://doi.org/10.2139/ssrn.6468518 [6].

The closest published match of the project's *model*: make m time-dependent, keep u static. Details, from the preprint's own text:

- Field taxonomy: *stable* (DOB, sex; fixed m, u), *dynamic* (low-cardinality: marital status, ZIP; semi-Markov/CTMC transition matrix), and *semi-stable* (high-cardinality: last name, address; a Weibull survival $S(\Delta t)$ on the current value).

- The match probability for an attribute is (eq. 8 of the paper)

$$m^*(\Delta t) = S(\Delta t) \cdot m_{\text{same}} + [1 - S(\Delta t)] \cdot m_{\text{changed}}, \qquad m_{\text{changed}} \approx u_c, \tag{3}$$

  which, for $S(\Delta t) = e^{-\Delta t/\bar{\tau}}$, is precisely $m(\Delta t) = u + (m - u)\, e^{-\Delta t/\bar{\tau}}$ — the form already used in the project's design note.

- The weights are the FS form with the time-dependent m: $w_{\text{agree}} = \ln(m(\Delta t)/u)$ and $w_{\text{disagree}} = \ln\big((1 - m(\Delta t))/(1 - u)\big)$.

- Experiments: NC Voter (10-year gap): TimeLink F1 0.877 vs. Splink 0.751; IPUMS historical census (20-year gap): 0.668 vs. 0.146; a synthetic 10–30 year study shows classical FS collapsing to 0.055 while TimeLink stays above 0.85.

Positioning relative to TimeLink: both make m decayable and u static; TimeLink derives $m(\Delta t)$ from generative sojourn-time models and precomputed lookup tables, while the present project proposes an additive pair-conditional term on the logged $\log_2(m_c/u_c)$ stack that composes with any m/u calibration, joined with measured per-field rates on a FAISS/Splink pipeline — a measurement, not just a model.

3 The entropy reading of decay

The literature as summarized converges on "transient fields decay, stable fields do not" but leaves the form and the benefit-conditional evidence open. The present section records an information-theoretic view that unifies these observations.

- **FS weights are information in bits.** Each contribution in (1) is a log-likelihood-ratio: its match-conditional expectation is the Kullback–Leibler divergence

$$\mathbb{E}\left[\log_2 \frac{P(\gamma \mid M)}{P(\gamma \mid U)}\right] \;=\; \sum_{\gamma} P(\gamma \mid M) \log_2 \frac{P(\gamma \mid M)}{P(\gamma \mid U)} \;=\; D_{\mathrm{KL}}(M_c \| U_c) \;\geq\; 0, \tag{4}$$

  the expected "surprise" of seeing the pair (agree vs. disagree) in the match vs. chance case." A decayed weight of 0 carries no information: the outcome is equally expected under M and U.

- **u_c is a collision (Rényi) entropy.** In a frequency model, $u_c(v) = f_c(v)$, the value frequency. The probability of a random coincidental agreement is the quadratic collision $\sum_v f(v)^2 = \exp(-H_2)$ with H_2 the Rényi-2 entropy of the value distribution. So "common value vs. rare value" is exactly a difference of H_2: a rare agreement carries more surprise — $-\log_2 u$ worth — the entropy content of the field.

- **Decay erases mutual information.** Under the time-dependent model $P(\cdot \mid M, \Delta t) \to P(\cdot \mid U)$, the KL diverges to zero and the pointwise weights go to zero; the mutual information $I(\gamma; M \mid \Delta t)$ between the comparison outcome and the match indicator falls to zero as the gap ages. For the exponential $m(\Delta t) = u + (m - u)e^{-\Delta t/T}$, the pointwise weight halves after about $T \ln 2$.

- **Noise entropy, not loss of the data.** The decay raises the match conditional entropy $H(P(\cdot \mid M, \Delta t))$ towards $H(P(\cdot \mid U))$: the observed contact fields of a true pair become as uninformative as the fields of an arbitrary pair. Decay therefore is a statement about the *expected entropy of the evidentiary outcome*, not about the information being lost from the data set — the surname is still itself; it is just no longer evidence.

Crucially, entropy is concerned with *how much* information, not its *direction*; the caveats recorded for the temporal literature above are not alternatives to this reading. The entropy framing explains the sign of a known phenomenon — why pairs at long gaps should trust identity anchors and discount transient contact fields — and provides per-field "half-life" numbers that are directly measurable from a labelled corpus, which is what the measured experiments in the present project attempt.

4 Against the "ad hoc decay function" objection

TimeLink's central methodological complaint about earlier temporal work is that it "relies on ad hoc decay functions rather than deriving weights from a coherent generative model." The entropy reading suggests that the decay is not a free knob to be pulled but the *only* information-theoretically justified completion of the static FS model once a field's lifetime enters. This section makes that precise.

Step 1: the weight keeps its likelihood-ratio form. A Fellegi–Sunter weight is a Bayes factor, and Bayes factors are not chosen: they are determined by the model for the comparison outcome. Adding the capture-date gap Δt as a covariate does not change the decision rule — the optimal evidence for deciding M versus U given γ and Δt is always the ratio

$$w_c(\ell; \Delta t) \;=\; \log_2 \frac{P(\ell \mid M, \Delta t)}{P(\ell \mid U, \Delta t)}. \tag{5}$$

There is no alternative weight family: any other scalar evidence leaves discrimination bits on the table (Neyman–Pearson optimality). So "decay" cannot be introduced as a new modeling act; it is forced once the model conditions on Δt.

Step 2: decompose into a baseline and a max-entropy survival part. Write the match-conditional agreement probability as

$$m_c(\Delta t) \;=\; u_c + (m_c(0) - u_c)\, S_c(\Delta t), \tag{6}$$

where $S_c(\Delta t) \in [0, 1]$, $S_c(0) = 1$, is the survival of the field's current value, and $m_c(0)$ is its modern agreement probability. The static model is the special case $S_c \equiv 1$. Which functional form for S_c? The information-theoretic default is *maximal ignorance*: given only the mean residency $\bar{\tau}_c$ of a field's value and no other structural knowledge, the entropy-maximising distribution for a positive survival time is the exponential $S_c(\Delta t) = e^{-\Delta t/\bar{\tau}}$, so that

$$m_c(\Delta t) \;=\; u_c + (m_c(0) - u_c)\, e^{-\Delta t/\bar{\tau}_c}, \tag{7}$$

which is (6) with the entropy-maximising survival. If the variance (or shape) of residency is empirically available, the max-entropy form becomes the Weibull $S_c(\Delta t) = e^{-(\Delta t/\bar{\tau})^k}$— the two-parameter generalisation that TimeLink selects too, on generative grounds rather than entropy grounds. The "decay rate" $\rho_c = 1/\bar{\tau}_c$ is therefore not a knob tuning a weight; it is the parameter of the max-entropy survival of the field's residency distribution, estimable from labelled history alone.

Step 3: estimation from data, not assumption. Both $m_c(0)$ and $\bar{\tau}_c$ are estimable (e.g. maximum-likelihood on the observed gaps of matched pairs: $\bar{\tau}_c$ is the empirical mean residency of changed values, and $m_c(0)$ the same-pair agreement at near-zero gap). The decay is then *not* *assumed to be a specific multiple of the static weight; it is inferred.* The general form (5) specialises, with (7), to

$$w_c(\Delta t) = \log_2 \frac{u_c + (m_c(0) - u_c)e^{-\Delta t/\bar{\tau}_c}}{u_c}, \tag{8}$$

and the "decay multiplier" $\beta_c(\Delta t) = \log_2 m_c(\Delta t)/u_c$ falls from $\log_2(m_c(0)/u_c)$ to 0 at rate $\bar{\tau}_c^{-1}$, exactly as the observed agreement distributions prescribe.

What this says about the TimeLink comparison. TimeLink is right that a survival form and a decay curve must be identified to leave the static model. But "ad hoc" is not the correct characterisation of the exponential form in the present investigation: with only a mean field lifetime available, the *maximal-entropy* survival is exponential, and the empirical-fit procedure (mean survival time on actual matched pairs) identifies both parameters without appeal to a specific functional form. TimeLink and the design of this project differ in what they fit (full semi-Markov transition vs. per-field exponential periods), but both place the decay inside m, both learn the parameters from data, and neither requires an external knob. Where the present project goes beyond even TimeLink is Section 5, item 1: the fitted $\bar{\tau}_c$ is reported in half-lives, and those half-lives predict (and are checked against) the measured $D_{KL}(\Delta t)$ curves of each field.

Caveat: entropy vs. decision. Bayes factors that take entropy-compat form must still be used with a decision rule. Entropy tells us what the evidence is (bits, sign preserve the log odds); it does not select the match prior λ or the threshold τ, which remain decision-theoretic choices (the costs of FP and FN). Nothing in the max-entropy derivation removes those. If anything, it separates them cleanly: the *model* is fixed by max-entropy-evidence, the *operating point* by cost.

5 Gap analysis

To the best of our knowledge:

1. no published work measures per-field decay rates (half-lives) on a two-stage embedding-then-linkage pipeline and relates them both to blocking recall and to the entropy framing above;
2. nobody has formulated the natural blocker consequence — drop a field from a fused index once it has roughly exhausted its mutual information — although TimeLink's field-specific lookups touch on this idea;
3. the "u static" assumption of the time-dependent-FS line is not carefully tested against *cohort drift* (the value distribution itself changes over decades, changing u), a concern present in Hu et al. (2017), in the Li et al. (2011) frequency-aware rates, and in the present project's design note;
4. the practical weak spot remains the NC Voter file, on which temporal models are observed to be fragile (Christen–Gayler 2013; Hu et al. 2017) — exactly where the project's own runs contribute additional evidence.

6 Notes on source status

This survey was compiled from public records (VLDB, ACM, Springer, OpenAlex, and DBLP) and from the accepted manuscripts of Christen–Gayler 2013 and Hu et al. 2017 where they could be read in full. TimeLink (a 2026 SSRN preprint) is cited from its abstract and any accessible body text; it was not wholly available to the survey. Where page-level facts could not be verified in the primary text (figure numbers in Christen–Gayler 2013), the survey says so explicitly. This review is part of the ongoing entity-resolution project, with the purpose of positioning the "measured per-field decay rates" result.

References

[1] I. P. Fellegi and A. B. Sunter. A theory for record linkage. JASA 64(328):1183–1210, 1969.

[2] P. Li, X. L. Dong, A. Maurino, and D. Srivastava. Linking temporal records. PVLDB 4(11):956–967, 2011.

[3] P. Christen and R. W. Gayler. Adaptive temporal entity resolution on dynamic databases. PAKDD 2013, LNAI 7819:558–569.

[4] Y. Chiang, A. Doan, and J. F. Naughton. Modeling entity evolution for temporal record matching. SIGMOD 2014:1175–1186.

[5] Y. Hu, Q. Wang, D. Vatsalan, and P. Christen. Improving temporal record linkage using regression classification. PAKDD 2017, LNCS 10235:561–573.

[6] K. B. Shim. TimeLink: time-aware record linkage with semi-Markov dynamics. SSRN preprint 6468518, 2026.

[7] W. E. Winkler. String comparator metrics and enhanced decision rules in the Fellegi–Sunter model of record linkage. Proceedings of the American Statistical Association Section on Survey Research, 1993.